



Cyber Incident Response - Playbooks

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Agenda

1. Cybersecurity foundations: CIA Triad
2. Understanding attack progression: Cyber Kill Chain
3. NIST CSF and the holistic incident-response view
4. Cyber incident-response operations
5. Blue Team response and recovery process (NIST SP 800-61)
6. Incident detection, analysis, triage, and communications
7. Major vs minor incidents and IT/OT change management
8. Incident-response strategies across IT and OT
9. Major OT incident response and recovery playbooks

What is at Stake?



Confidentiality: Steal sensitive information through unauthorized access



Integrity: Compromise Data Integrity and Systems Trustworthiness

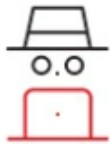


Availability: Systems not accessible to users/organizations



Cyber incidents impact one or more CIA principles

How Attackers Do It? Cyber Kill Chain



Reconnaissance

Attackers gather information on the target, such as open ports or employee emails.



Delivery

Sending the payload, typically via phishing emails or drive-by downloads.



Installation

Malware establishes persistence by installing backdoors or trojans.



Actions on Objectives

They achieve their goal, whether stealing data, encrypting files, or disrupting services.

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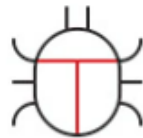
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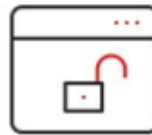
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Weaponization

They prepare malware payloads, often tying exploits to malicious files or links.



Exploitation

The malicious code runs on the target system, exploiting a vulnerability.



Command and Control (C2)

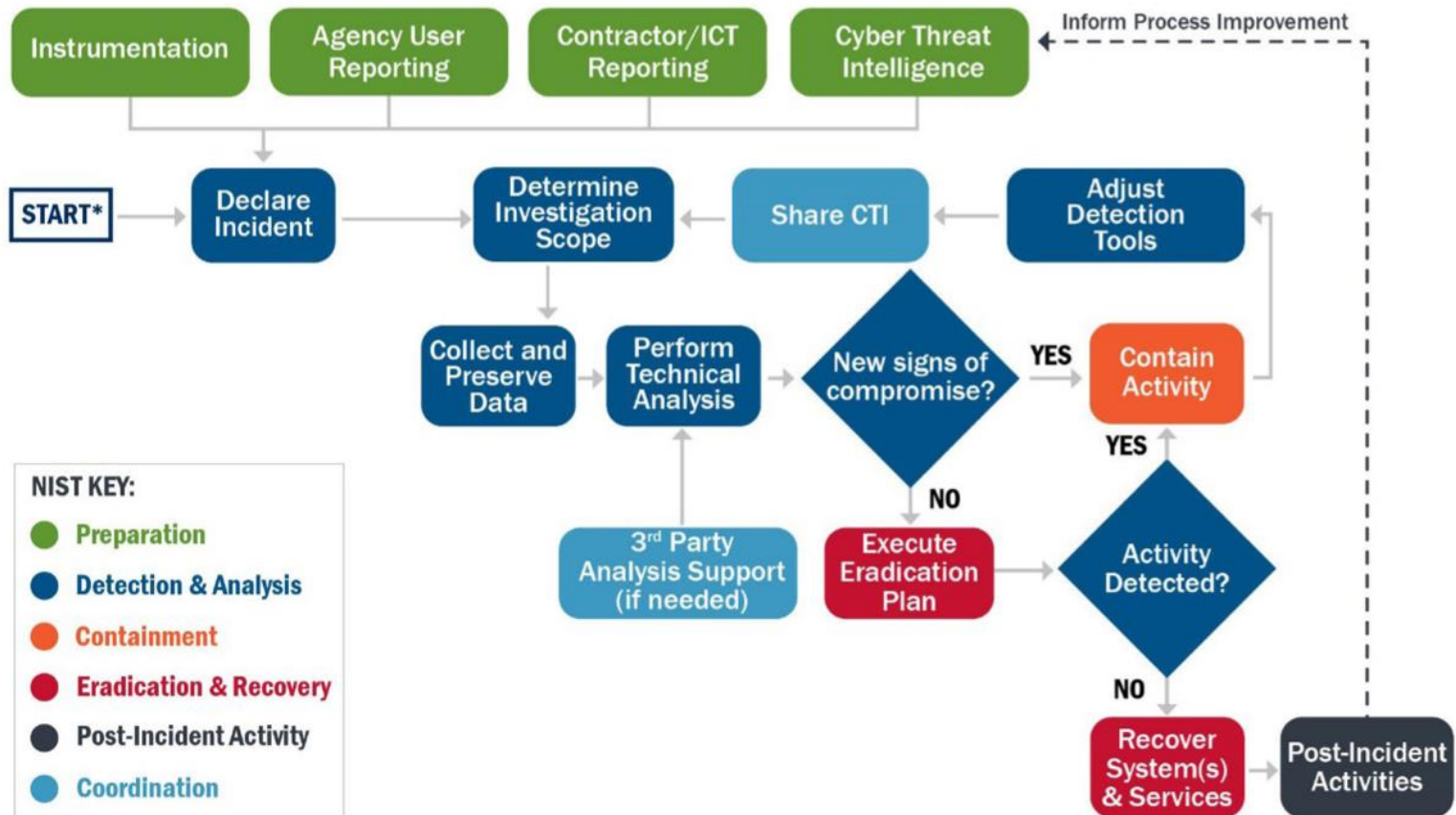
Attackers communicate with the compromised system to issue commands.

Towards Incident Response: A Holistic View

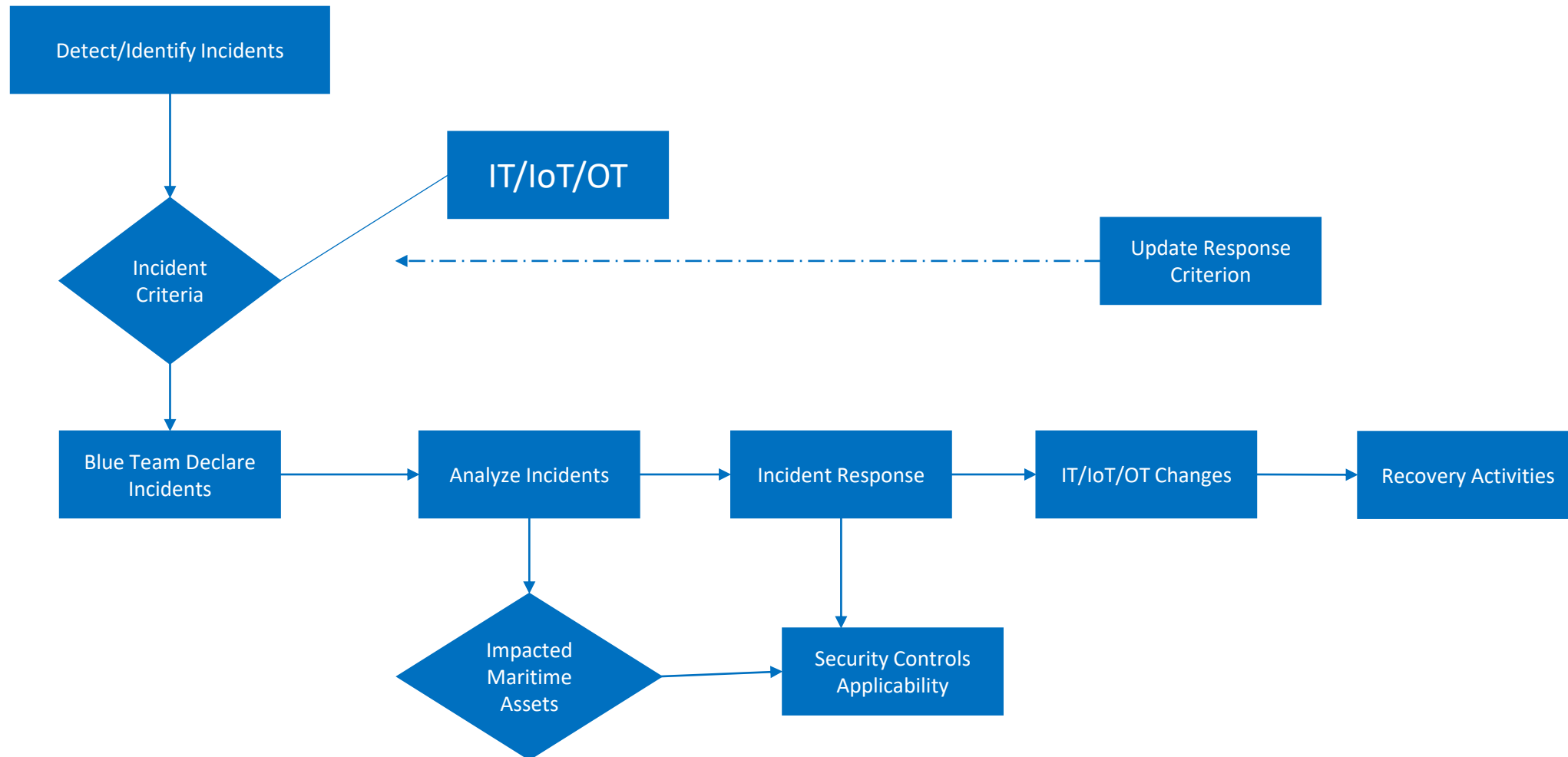
NIST Cyber Security Framework



Cyber Incident Response Operations



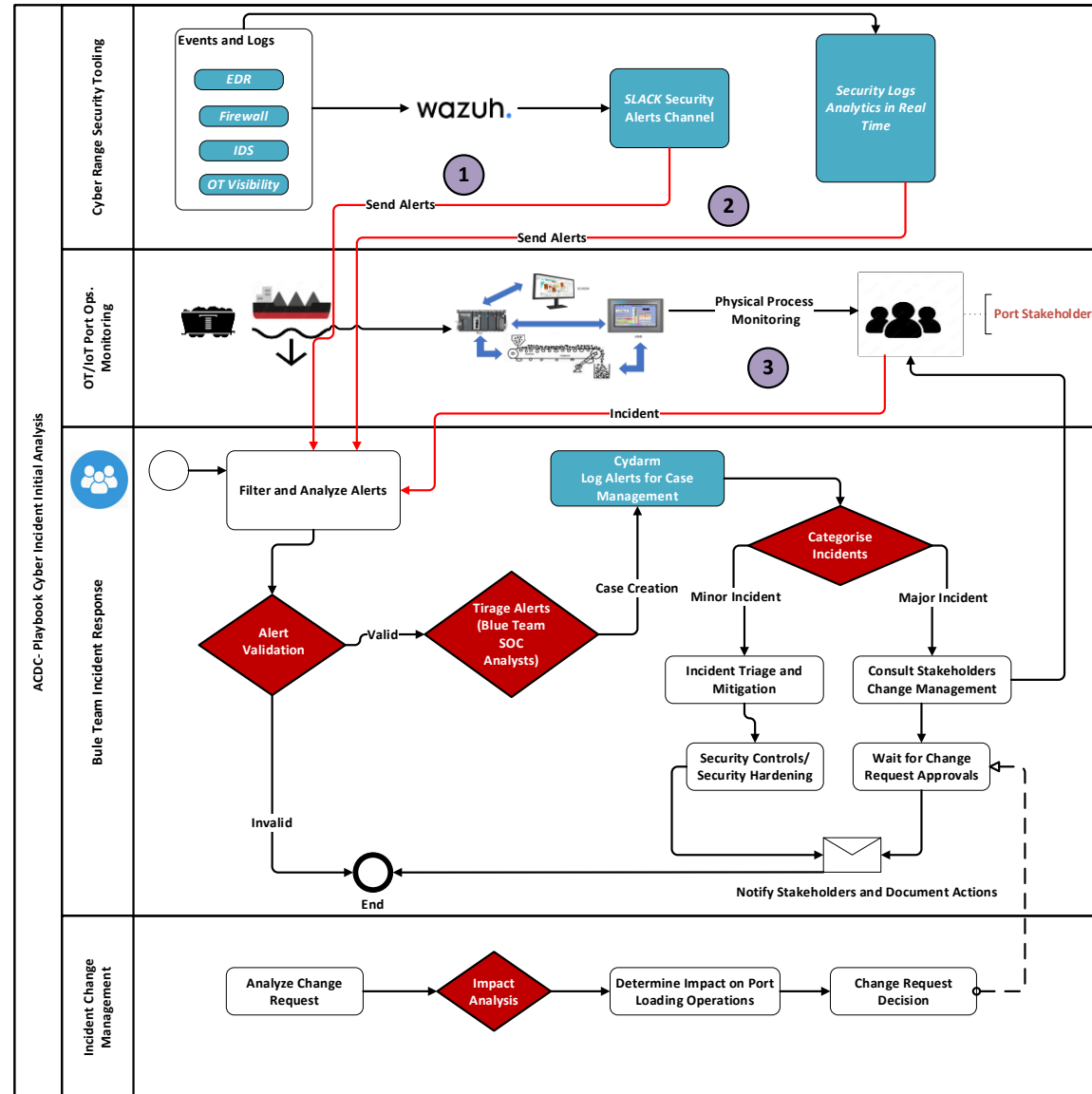
Blue Team Response and Recovery Process (NIST-SP 800-61)



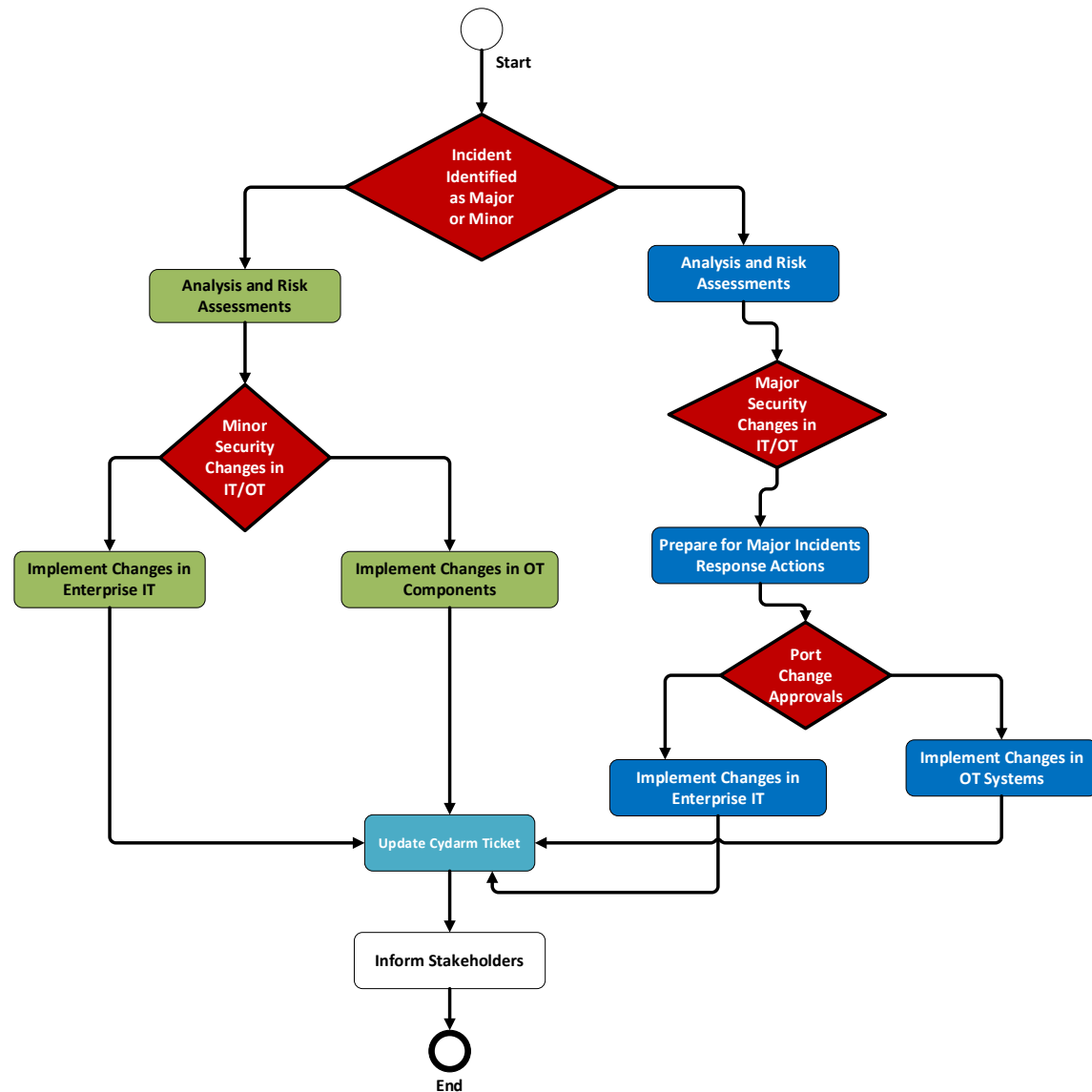
Blue Team – Incident Response Phases

- Incident Detection
- Incident Analysis
 - Minor Incidents
 - Major Incidents
- Communications

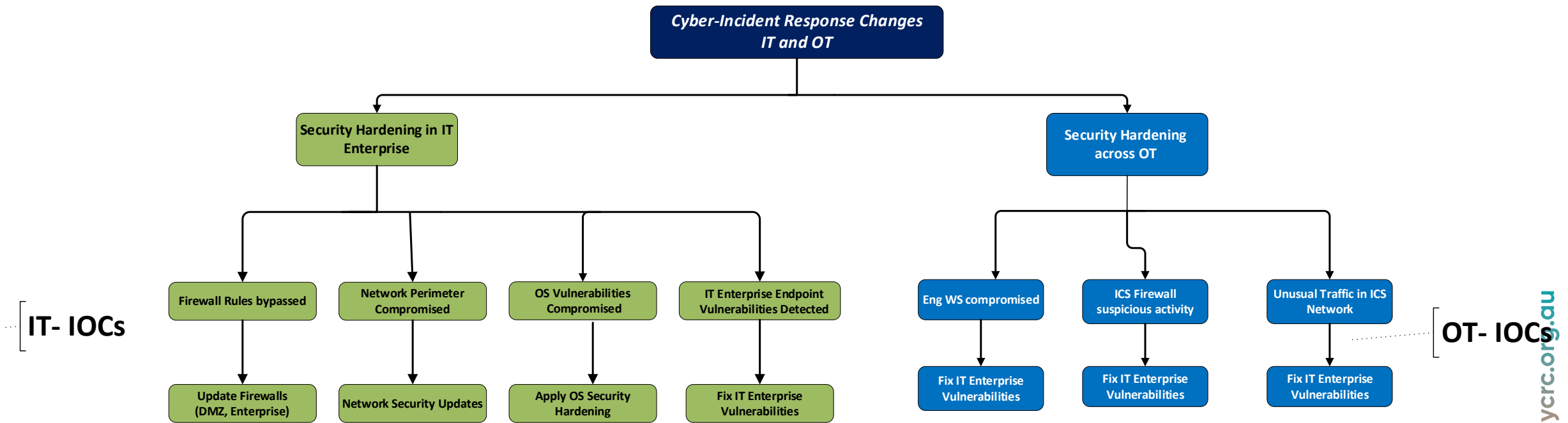
Blue Team – Incident Detection and Analysis



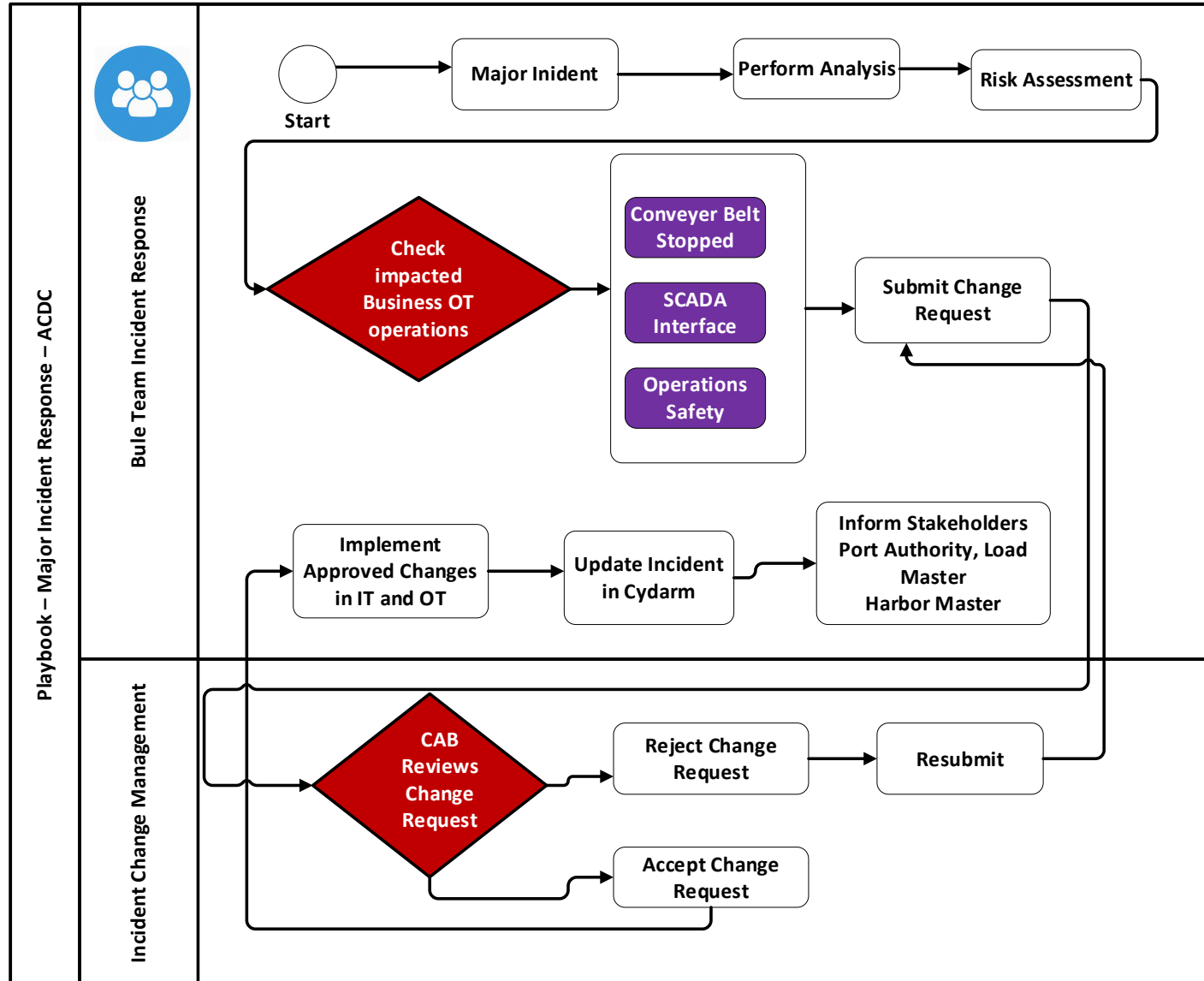
Blue Team – Navigating through Major and Minor Incidents



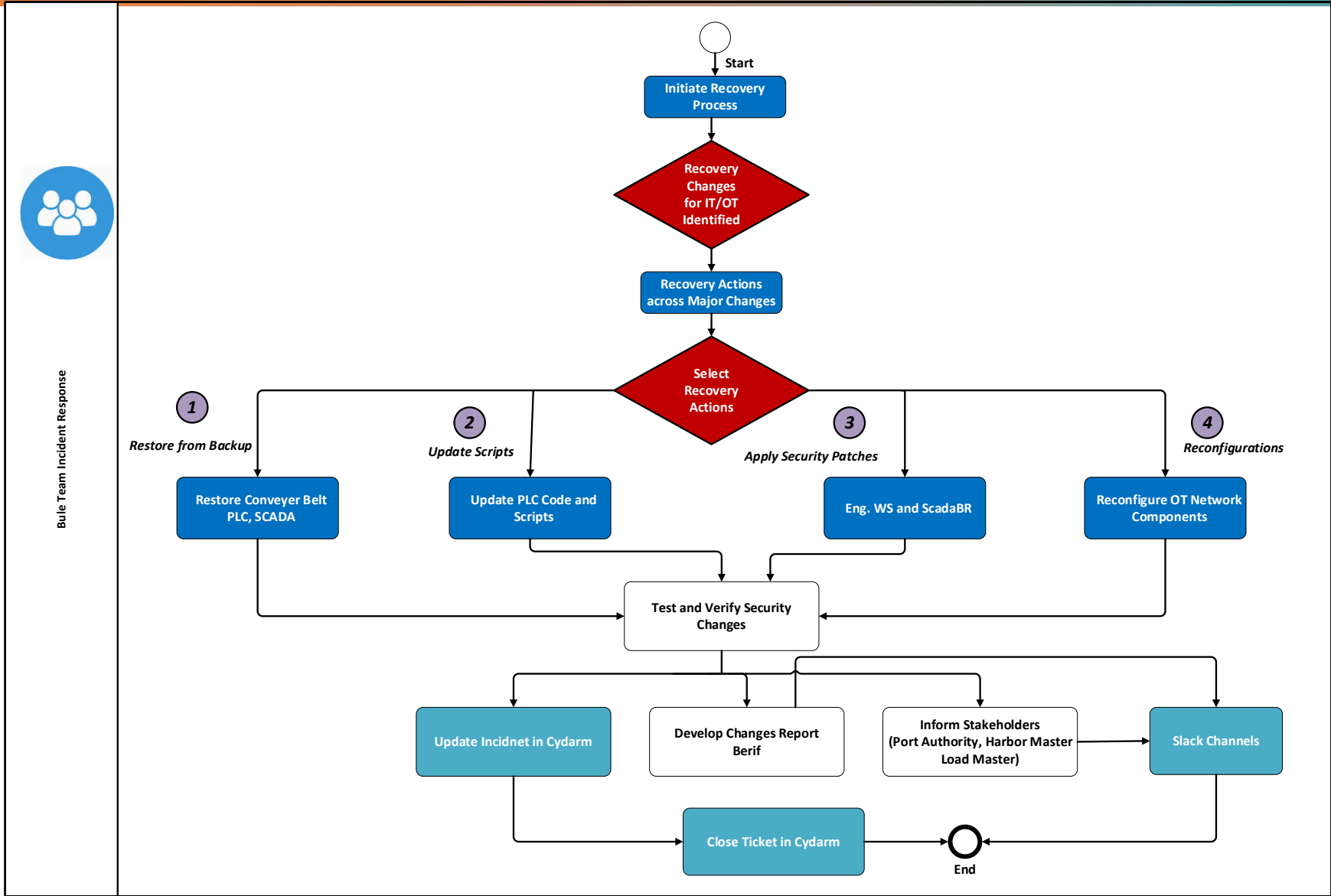
Incident Response Strategies



Blue Team – Major Incident Response for OT



Blue Team – Major Incident Recovery for OT



Thanks!